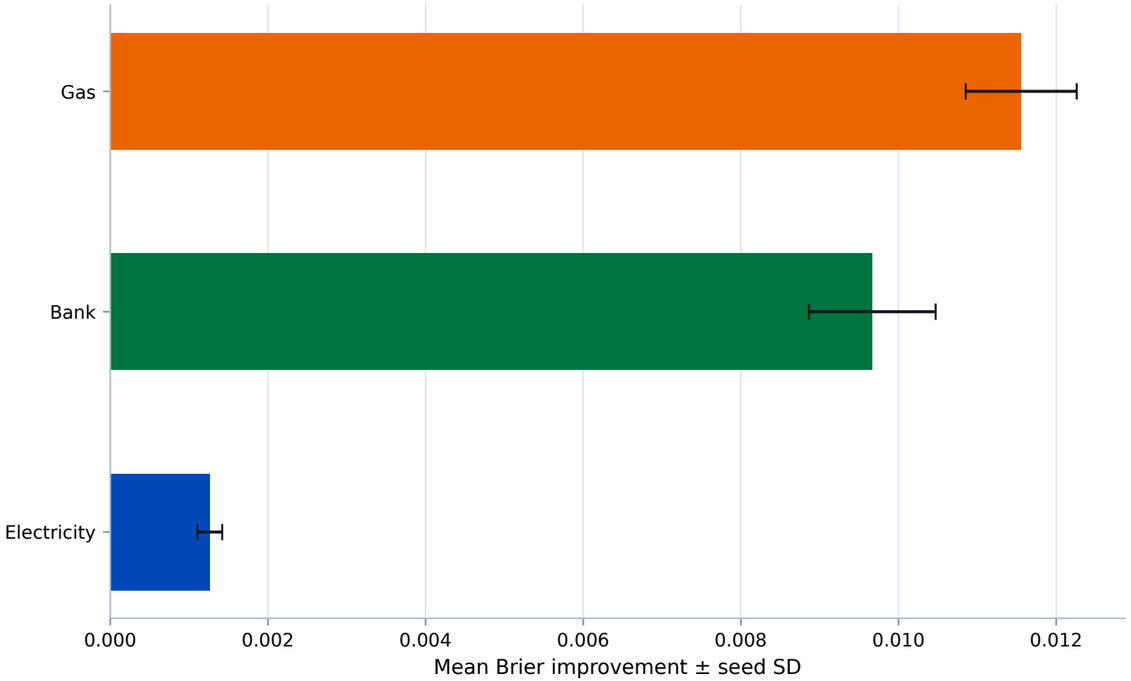
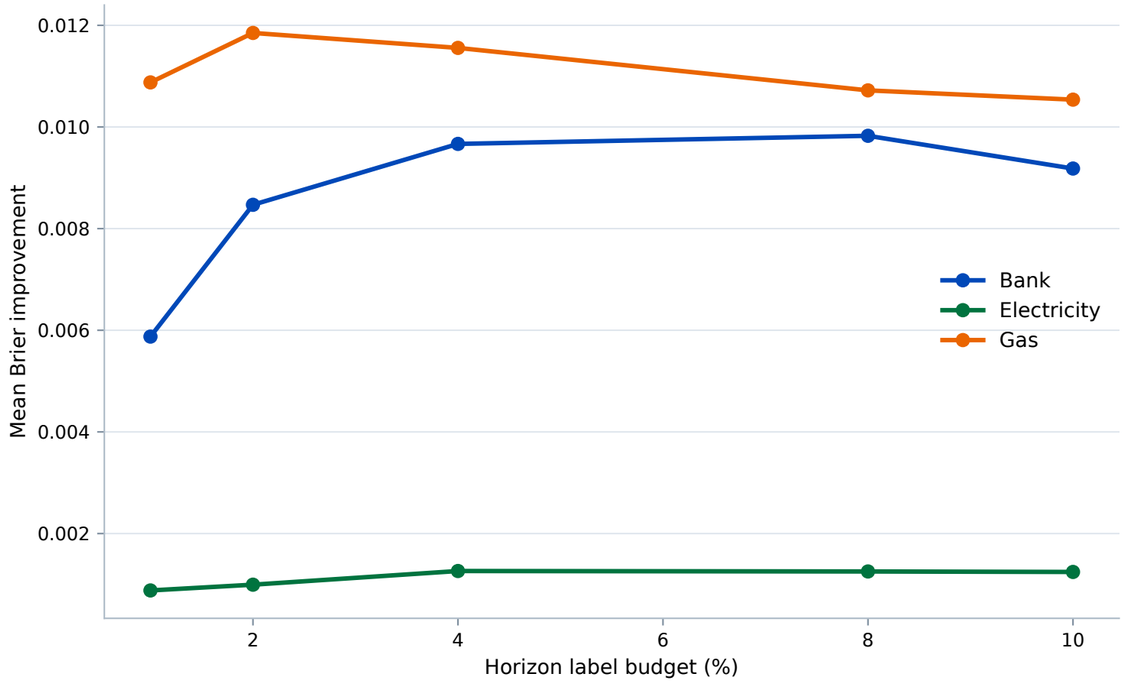
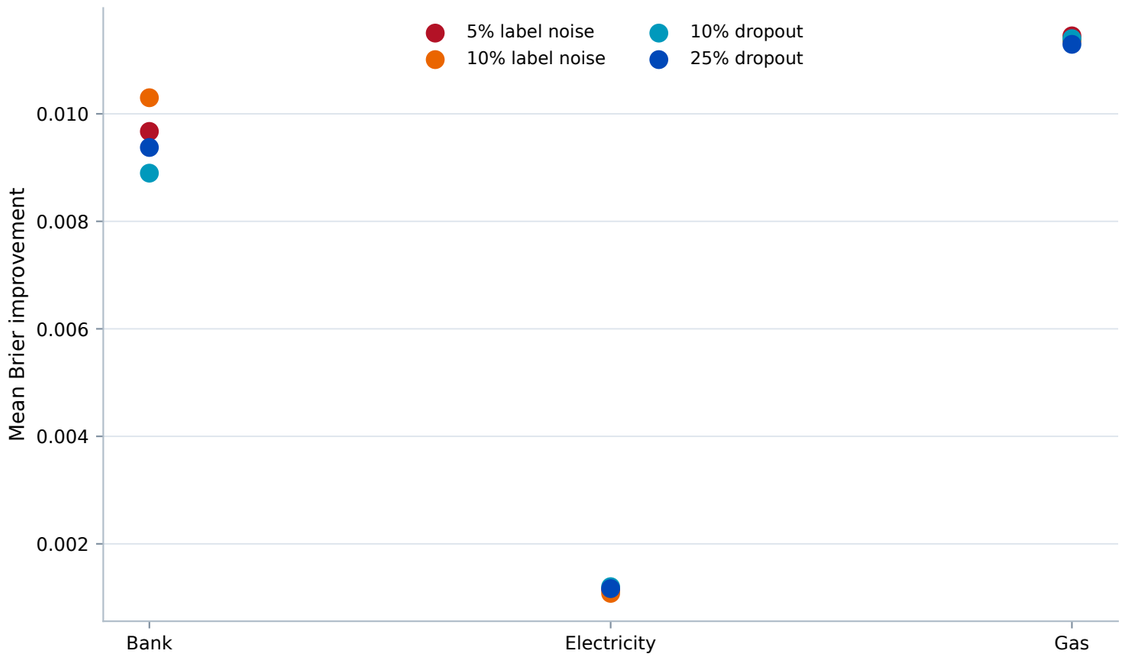
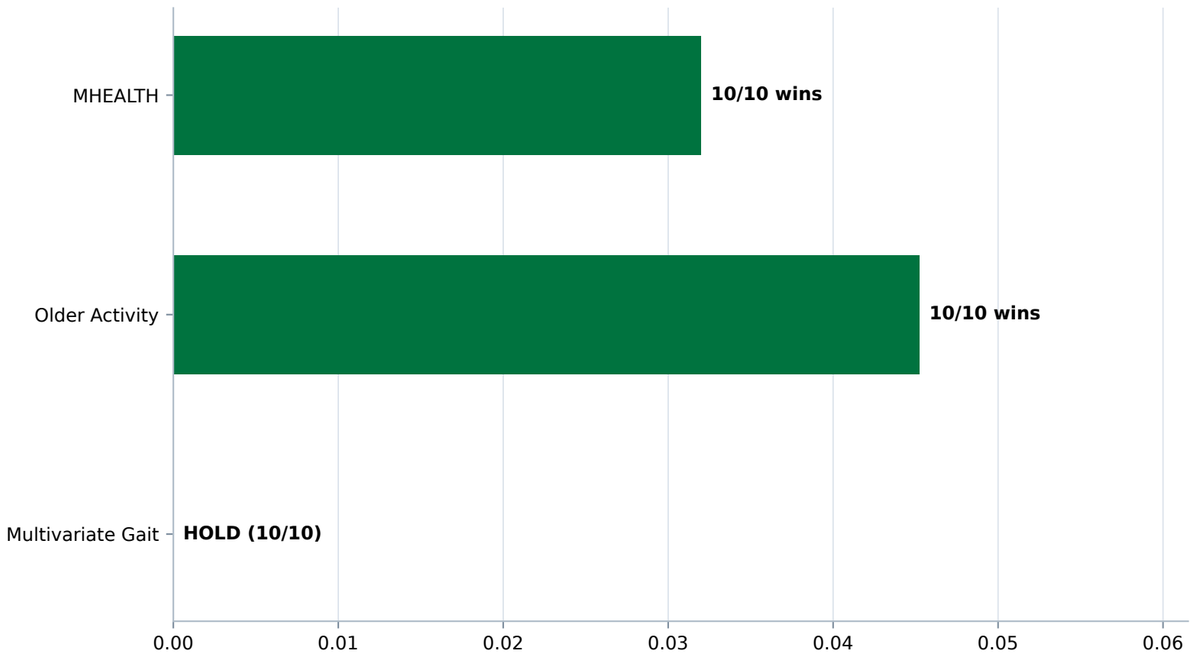
(a) Core sparse-maintenance outcomes



(b) Feedback-budget frontier



(c) Noise and canary-dropout stress



(d) Locked fresh confirmation and no-op control